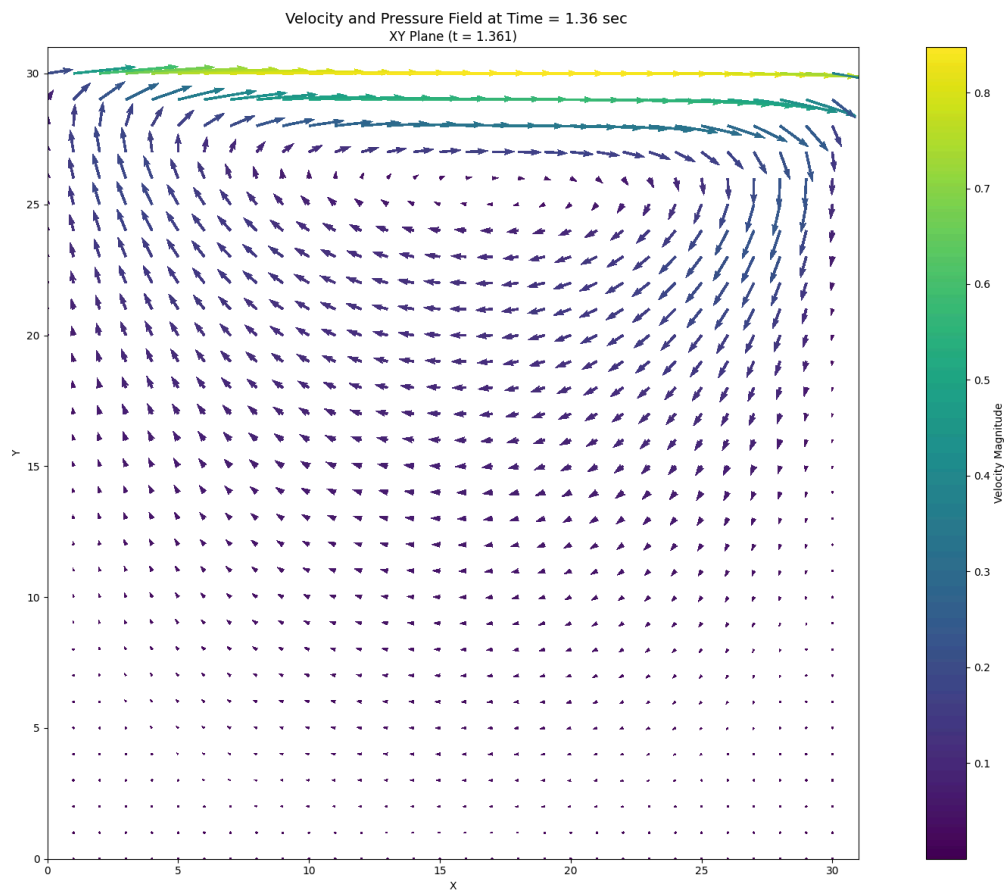


Experimental Report

Case: Poiseuille Flow with Moving Walls

Description of Flow: This case models the standard two-dimensional lid-driven cavity, in which the top wall moves tangentially while the remaining three walls remain stationary. The imposed lid motion generates a primary circulating vortex and weaker secondary recirculation regions, creating a strongly nonlinear velocity field. The experiment aims to recover the streamwise momentum equation from the recorded velocity and pressure data.

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

Learned equation

$$u_t = -1.13842289e+00*uu_x - 1.03055812e+00*vu_y + 8.40649079e-03*u_{xx} + 1.03461437e-02*u_{yy} - 1.04792514e-03*p_x$$

Term	Reference coefficient	Learned coefficient	Absolute error
uu_x	-1.00000000e+00	-1.13842289e+00	1.38422889e-01
vu_y	-1.00000000e+00	-1.03055812e+00	3.05581219e-02
u_xx	+1.00000000e-02	+8.40649079e-03	1.59350921e-03
u_yy	+1.00000000e-02	+1.03461437e-02	3.46143665e-04
p_x	-1.00000000e-03	-1.04792514e-03	4.79251383e-05

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	1.00000000e+03	kg m ^{-3}
Dynamic viscosity	μ	1.00000000e+01	Pa s
Kinematic viscosity	ν	1.00000000e-02	m ² s ^{-1}
Characteristic velocity	U	1.00000000e+00	m s ^{-1}
Characteristic length	D	1.00000000e+00	m
Reynolds number	$Re = \rho U D / \mu$	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s ^{-2}

3. Numerical Discretisation

Quantity	Symbol	Value	Unit
Grid resolution	$N_x \times N_y$	31 x 31	cells
Grid spacing	$\Delta x, \Delta y$	3.22580645e-02, 3.22580645e-02	m
Time step	Δt	1.00000000e-04	s
Numerical threshold	ϵ	7.00000000e-11	-
Advective coefficient	c_a	1.00000000e+00	-
Pressure coefficient	$c_p = 1/\rho$	1.00000000e-03	m ³ kg ^{-1}
Viscous coefficient	$c_v = 1/Re$	1.00000000e-02	-

4. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(6655, 30, 31)
Regression field shape (t, x, y)	(4255, 22, 27)

Quantity	Value
Regression samples	2,527,470
Dictionary terms	5
First saved time index	150
Temporal crop at each boundary	1200 layers
x-boundary crop	4 points
y-boundary crop	2 points

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	9.42668455e-04
L0 penalty	1.00000000e-06
Training RMSE	1.10494359e-02
Validation RMSE	6.16076275e-03